



Public Competition Assessment

10 January 2007

Hexion Specialty Chemicals Pty Ltd - proposed acquisition of Orica Australia Limited's formaldehyde resins business

Introduction

1. On 6 December 2006, the Australian Competition and Consumer Commission (the **ACCC**) announced its decision to not oppose the proposed acquisition of Orica Australia Limited's formaldehyde resins business by Hexion Specialty Chemicals Pty Ltd (**Proposed Acquisition**). The ACCC decided that the Proposed Acquisition would not have the effect, or be likely to have the effect, of substantially lessening competition in contravention of section 50 of the *Trade Practices Act 1974* (the **Act**) in the relevant regional market (including eastern South Australia, Victoria and extending to central New South Wales) for the manufacture and supply of amino resins, or in the national market for the manufacture and supply of phenolic resins.
2. The ACCC made its decision on the basis of the information provided by the merger parties and information arising from its market inquiries. This Public Competition Assessment outlines the basis on which the ACCC has reached its decision on the proposed transaction, subject to confidentiality considerations.

Public Competition Assessment

3. To provide an enhanced level of transparency and procedural fairness in its decision making process, the ACCC issues a Public Competition Assessment for all transaction proposals where:
 - a merger is rejected;
 - a merger is subject to enforceable undertakings;
 - the merger parties seek such disclosure; or
 - a merger is approved but raises important issues that the ACCC considers should be made public.
4. In this case, the ACCC considers the merger raises important issues, given the high level of market concentration in each of the relevant markets for the manufacture and supply of amino and phenolic resins.
5. By issuing Public Competition Assessments, the ACCC aims to provide the market with a better understanding of the ACCC's analysis of various markets and

the associated merger and competition issues. It also alerts the market to the circumstances where the ACCC's assessment of the competition conditions in particular markets is changing, or likely to change, because of developments.

6. Each Public Competition Assessment is specific to the particular transaction under review by the ACCC. While some transaction proposals may involve the same or related markets, it should not be assumed that the analysis and decision outlined in one Public Competition Assessment will be conclusive of the ACCC's view in respect of other transaction proposals, as each matter will be considered on its own merits.
7. Many of the ACCC's decisions will involve consideration of both non-confidential and confidential information provided by the merger parties and market participants. In order to maintain the confidentiality of particular information, Public Competition Assessments do not contain any confidential information or its sources. While the ACCC aims to provide an appropriately detailed explanation of the basis for the ACCC decision, where this is not possible, maintaining confidentiality will be the ACCC's paramount concern. Accordingly, a Public Competition Assessment may not definitively explain all issues and the ACCC's analysis of such issues.

The parties

*The acquirer: Hexion Specialty Chemicals Pty Ltd (**Hexion**)*

8. Hexion is headquartered in Ohio, USA and has significant experience in the production of formaldehyde based resins. In Australia, Hexion manufactures amino and phenolic resin from production plants in Laverton, Victoria and Brisbane, Queensland. The resins are primarily used in the manufacture of wood based products (eg particle board, MDF, plywood) and to a lesser degree in the production of foundry and glass fibre insulation.

*The target: Orica Australia Limited's (**Orica**) formaldehyde resins business*

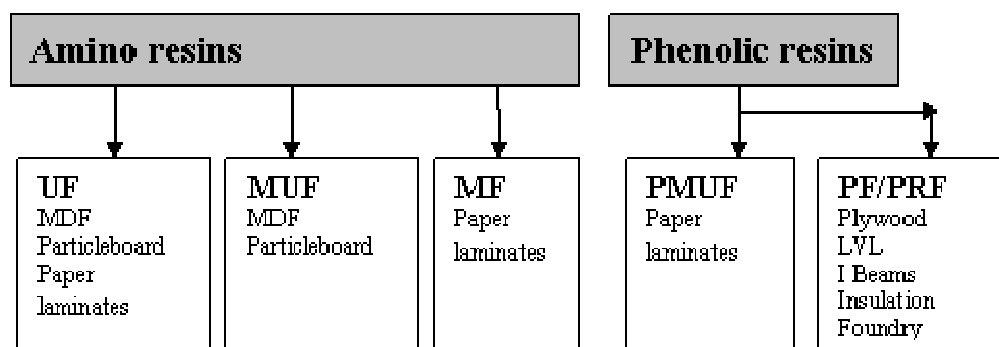
9. Orica is a publicly listed company on the Australian Stock Exchange. In Australia, Orica's formaldehyde resins business manufactures amino and phenolic resin from manufacturing facilities located at Deer Park in Victoria. Orica manufactures and supplies amino and phenolic resin products to the wood panels, timber, foundry and fibre insulation industries.

Market background

10. Formaldehyde resins come in a variety of forms to suit different applications. Broadly these different varieties of resin can be placed into one of two groups:
 - Amino resins – which includes urea formaldehyde (UF); melamine urea formaldehyde (MUF) and melamine formaldehyde (MF); and
 - Phenolic resins – which includes phenol melamine urea formaldehyde (PMUF*) and phenol formaldehyde (PF/PRF).

*PMUF is a hybrid amino/phenolic resin product.

11. Resins are adhesive agents that have a variety of applications, particularly in the wood based products industry and the foundry and fibreglass industries.
12. In Australia, Hexion and Orica manufacture and supply both amino and phenolic resins to customers who are typically manufacturers of wood based products such as particleboard, paper laminates, plywood, laminated veneer lumber, engineered lumber and structural beams. The supply of resin typically involves the provision of additional on-site or local after sales services and technical support to customers to ensure compatibility of the resin to raw wood materials which can vary by region and by season.
13. The diagram below illustrates the end uses of amino and phenolic resins.



14. In Australia, Hexion and Orica are the only manufacturers of phenolic resins. There are also limited volumes of imported phenolic resins.
15. Hexion and Orica also manufacture amino resins. Amino resins are also manufactured in Australia by other independent manufacturers of amino resin - Woodchem (Oberon, central New South Wales) and Dynea (Dardanup, Western Australia).

Timing

16. The following table outlines the timeline of key events in this matter.

Date	Event
1 st September 2006	ACCC commenced review under the Merger Review Process Guidelines. Market inquiries commenced.
15 th September 2006	Closing date for submissions from interested parties.
6 th October 2006	Having regard to the submissions received, the proposed date for ACCC findings was revised from 13 October to 18 October.
10 th October 2006	Proposed announcement delayed until 25 October (delay requested by Hexion to allow for provision of additional information).
18 th October 2006	Proposed announcement delayed to allow for further inquiries.
1 st November 2006	ACCC published a Statement of Issues outlining preliminary competition concerns. Proposed date for announcement of ACCC findings delayed to allow for market consultation on the ACCC's Statement of Issues.
15 th November 2006	Closing date for submissions from interested parties.
6 th December 2006	ACCC decided not to oppose the proposed acquisition.

Market inquiries

17. The ACCC conducted market inquiries with a range of industry participants, including competitors, customers, relevant industry associations, other regulatory agencies and other interested parties.

Statement of Issues

18. On 1 November 2006, the ACCC published a Statement of Issues (SOI). In the SOI, the ACCC sought information and comment on a number of issues important to its assessment of the competition effects (if any) of the Proposed Acquisition. A number of these issues are set out below.

- The degree to which amino resin manufacturers in central New South Wales or other regions would be able to constrain the merged entity in the market for the manufacture and supply of amino resins.
- The likelihood and ability of large buyers of resins to sponsor entry into the manufacture of amino and/or phenolic resins. The ACCC sought information on the minimum level of demand required to justify new entry, the level of demand in relevant geographic regions, the availability of candidates for sponsorship etc.
- The likelihood that independent manufacturers of amino resins could expand into the manufacture of phenolic resins. The ACCC sought further information on how quickly such expansions could occur, the costs involved

and the circumstances under which independent manufacturers of amino resins would consider such an expansion.

- The likely competitive constraint on the merged entity from imports of phenolic resin, taking into account price competitiveness, shelf life and logistical issues.
- Constraints posed by competition in downstream markets. The ACCC sought information on the extent to which international competition faced by domestic producers of wood-based products (consumers of formaldehyde resins) would constrain the prices of amino and/or phenolic resins charged by the merged entity.

Market One: manufacture and supply of amino resins

Market definition

19. The ACCC considered amino resins and phenolic resins are currently supplied in separate product markets. This was based on the limited ability of major users of these resins to substitute from one to the other.
20. The ACCC's inquiries revealed that the relevant geographic market for amino resins was likely to be regional and to comprise eastern South Australia, Victoria, and extend to central New South Wales. The ACCC considered the viability of amino resin supply from central New South Wales to customers located in eastern South Australia taking into account factors such as price, freight costs and capacity. The ACCC considered that the supply or potential supply of amino resins from central New South Wales to eastern South Australia would be likely to constrain the merged entity. However more distant supply or potential supply would be unlikely to constrain the merged entity. The ACCC therefore considered that the geographic market for amino resins was likely to be regional and include eastern South Australia, Victoria and extend to central New South Wales.

Competition analysis

Market concentration

21. The ACCC found that the proposed acquisition would result in increased concentration in the regional market for amino resins with the number of players reduced from three to two.
22. However, the ACCC considered that Woodchem (located in Oberon, New South Wales) would remain as a competitive constraint on the merged entity. The ACCC considered that the potential for customers of Hexion or Orica to switch to Woodchem if the merged firm raised prices, including via additional transport costs from New South Wales to customers in South Australia, would be a viable constraint on the pricing of the merged entity. Further confidential information supported the ACCC's view in this regard.

Bargaining power

23. Information before the ACCC also revealed that customers in the relevant region would be likely to continue to possess a degree of bargaining power post acquisition. Among other things, the ACCC took into account the size of some amino resin customers taking into account their resin demand both within and outside the relevant geographic region.

Likelihood of sponsored entry

24. The ACCC's inquiries revealed some barriers to sponsored entry in the amino resins market in the relevant geographic region. These include the significant capital costs of plant establishment and the lack of demand to enable a new entrant to operate at an efficient scale of production.

Conclusion – Market One

25. The ACCC did not attach significant weight to any competitive constraint on the merged entity arising from the possibility of sponsored entry in the relevant region. However, the ACCC considered that the Proposed Acquisition would be unlikely to result in a substantial lessening of competition in this market on the basis of, among other things, the competitive constraint on the merged entity posed by Woodchem and the bargaining power of some customers.

Market Two: manufacture and supply of phenolic resins

Market definition

26. The ACCC considered the impact of the Proposed Acquisition on competition in a national market for the manufacture and supply of phenolic resins.

Competition analysis

Market concentration

27. The ACCC found that the Proposed Acquisition would result in the merged entity being the only manufacturer and supplier of phenolic resins in Australia.
28. However, as discussed below, the ACCC's inquiries revealed that barriers to expansion into phenolic resin manufacture by existing amino resin manufacturers appeared to be relatively low.

Barriers to expansion into phenolic resin manufacture

29. Information before the ACCC indicated that independent manufacturers of amino resins could expand into the manufacture of phenolic resins within relatively short timeframes with relatively low capital investment. The ACCC's inquiries also revealed that there appears to be sufficient levels of phenolic resin demand in Australia to justify expansion by existing amino resin manufacturers should customers wish to switch their source of phenolic resin supply.
30. The ACCC further noted the existence of two amino resin manufacturers in Australia, other than the merged entity, who could potentially expand into phenolic resin manufacture.
31. The ACCC considered that the threat of expansion by existing amino resin manufacturers was likely to be a competitive constraint on the merged entity.

Import competition

32. Information before the ACCC indicated that while there are imports of some phenolic resin into Australia, imports of the type of phenolic resin suitable for use in the wood products industry (particularly PF resin) have been limited.
33. The ACCC's inquiries revealed a number of impediments to the importation of phenolic resin suitable for use in the wood products industry including shelf life, logistics and price competitiveness taking into account transport costs or possible need for specialised shipping containers.

34. While imports of phenolic resin suitable for use in the wood products industry appeared to be technically feasible, the constraint on the merged entity posed by the threat of customers switching to imports was likely to be limited.

Competition in downstream markets

35. The ACCC's inquiries revealed that the indirect constraint on the merged entity from competition in downstream markets was unlikely to be significant. Information before the ACCC revealed that imports constitute approximately 50% of the plywood used in Australia. An argument put to the ACCC was that the merged entity supplies resins to domestic plywood producers, and an increase in the price of resins would make domestic producers of plywood less competitive; to the extent domestic producers of plywood lose customers to imported plywood, the merged entity may suffer consequently and this would be a constraint. However, on close examination, as phenolic resin represents a small proportion of the cost of producing plywood, in the ACCC's view the indirect constraint on the merged entity from imports of plywood was likely to be weak.

Conclusion – Market Two

36. The ACCC did not attach significant weight to any competitive constraint arising from import competition or competition in downstream markets. However, the ACCC considered that the Proposed Acquisition would be unlikely to result in a substantial lessening of competition in this market on the basis of, among other things, the competitive constraint on the merged entity posed by relatively low barriers to expansion into phenolic resin manufacture faced by existing independent manufacturers of amino resin.

Conclusion

37. On the basis of the above, and other factors that were unable to be referred to in this Public Competition Assessment due to the confidential nature of that information, the ACCC considered that the Proposed Acquisition would be unlikely to result in a substantial lessening of competition in the regional market (including eastern South Australia, Victoria and extending to central New South Wales) for the manufacture and supply of amino resins and the national market for the manufacture and supply of phenolic resins.